

- 2 Rice, *Oryza sativa*, is an important food crop. Rice plants are wind pollinated. Pollen containing the male gamete is transferred by the wind to female reproductive organs of rice plants. Fertilisation and grain formation then occur.

Weedy rice is a wild form of rice that grows in fields of cultivated rice. Weedy rice competes with cultivated rice for resources. Weedy rice plants are taller than cultivated rice plants and produce a low yield of rice grains.

Several genetically modified (GM) varieties of cultivated rice have been developed.

One concern about the use of GM rice is gene flow from GM rice plants to weedy rice. Gene flow occurs when the wind carries pollen from GM rice plants to weedy rice plants.

Some scientists investigated gene flow from herbicide-resistant GM rice plants to weedy rice plants. The GM rice plants had a gene for herbicide resistance. Weedy rice plants do not have the gene for herbicide resistance.

The scientists wanted to test the hypothesis that:

Gene flow from GM rice to weedy rice decreases as the distance between the GM rice crop and weedy rice increases.

The scientists planted GM rice and weedy rice in a field, as shown in Fig. 2.1.

Fig. 2.1 also shows that the wind normally blows in a north-west (NW) direction.

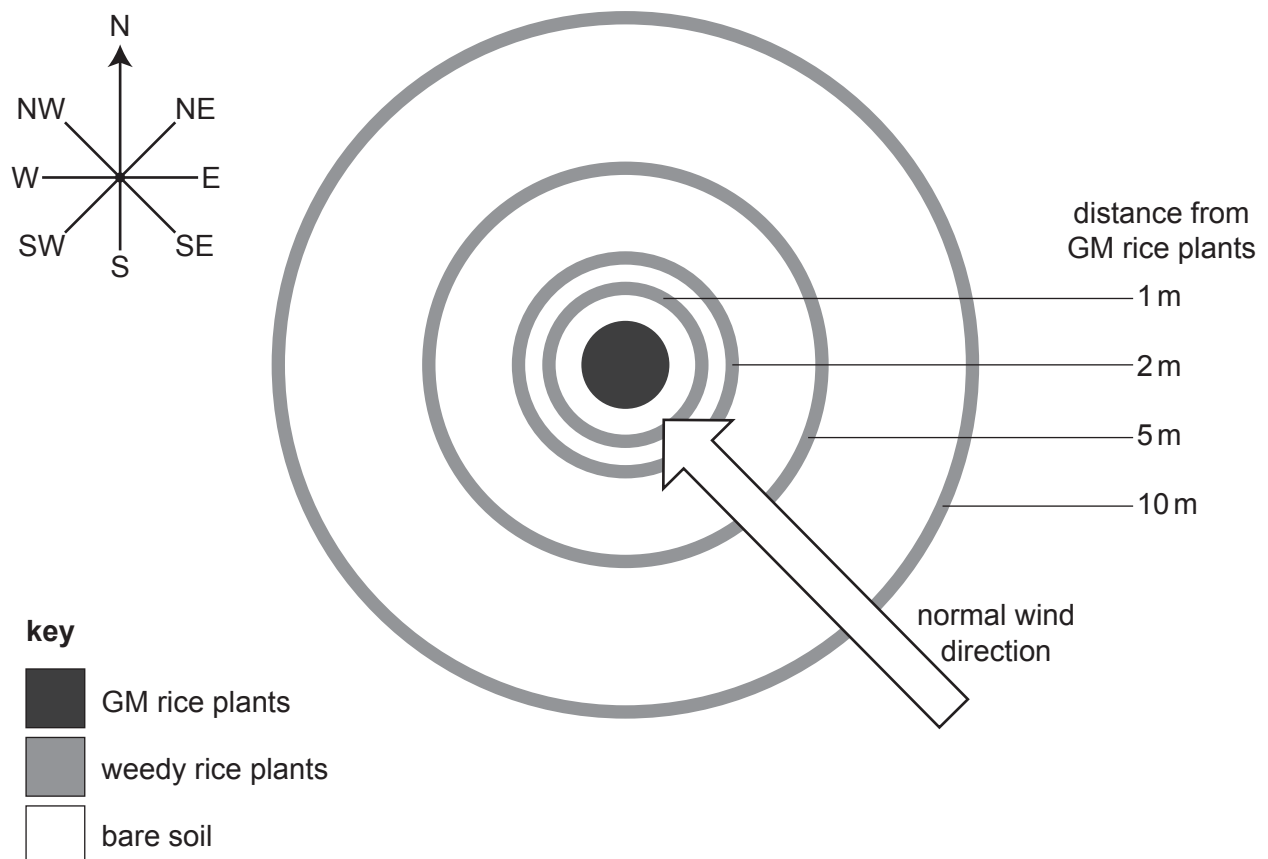


Fig. 2.1

After the plants had been pollinated and the rice grains had developed, the grains were collected from the **weedy** rice plants **only**.

- Grains were collected from all weedy rice plants growing 1 m from the GM rice plants in the directions N, NE, E, SE, S, SW, W and NW.
- Grains were also collected in the same directions from the GM rice plants at distances of 2m, 5m and 10m.
- Approximately 1000 grains from each collection point were planted and germinated in controlled conditions in a glasshouse.
- The plants were grown for three weeks and then tested to determine whether they had the gene for herbicide resistance.

Fig. 2.2 shows young rice plants three weeks after germination.

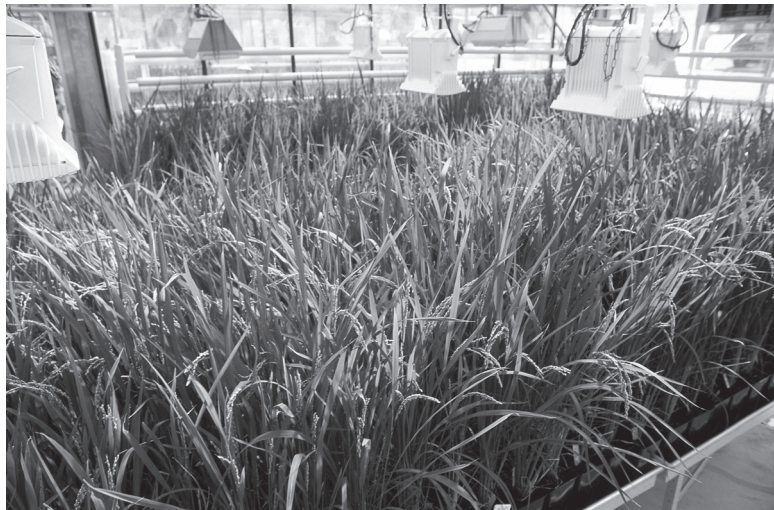


Fig. 2.2

(a) (i) Identify the **two** independent variables in this investigation.

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- (ii) Some of the young weedy rice plants had the gene for herbicide resistance. This gene had been carried to the parents of these plants in pollen from the GM rice plants.

Outline a method the scientists could use in the glasshouse to determine how many of the young weedy rice plants had the gene for herbicide resistance.

Your method should **not** require the extraction of nucleic acids.

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 [2]

- (iii) The scientists calculated the percentage of young weedy rice plants that had the gene for herbicide resistance. This percentage was used as a measure of gene flow.

The percentage gene flow recorded at each NW collection point is shown in Fig. 2.3.

Fig. 2.3 also shows the percentage gene flow recorded at each distance from the GM rice plants at all the other collection points combined.

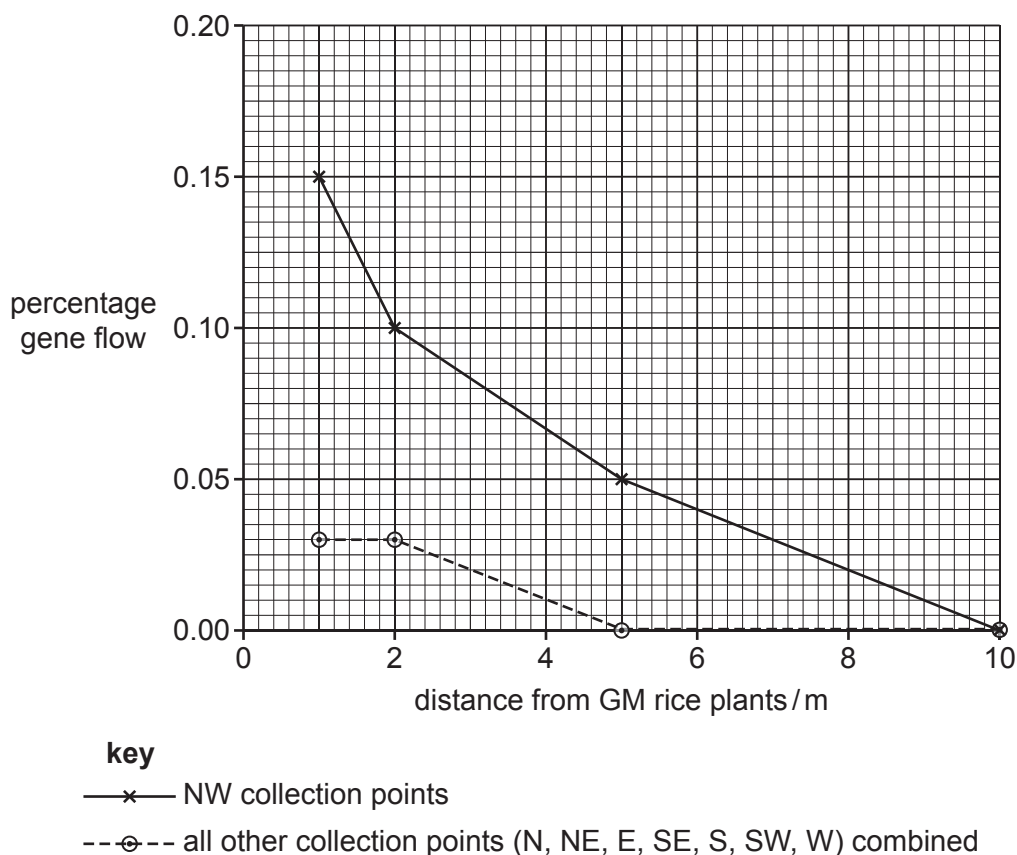


Fig. 2.3

State **two** conclusions that can be made from the data shown in Fig. 2.3.

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- (b) Weedy rice plants have a gene that results in increased height. GM rice plants do **not** have this gene. This gene allows gene flow from weedy rice plants to GM rice plants to be investigated.

In a second study, the scientists investigated gene flow from herbicide-resistant GM rice plants to weedy rice plants **and** gene flow from weedy rice plants to herbicide-resistant GM rice plants.

The scientists planted GM rice and weedy rice in areas next to each other in a field.

After the plants had been pollinated and the rice grains had developed, 1000 grains were collected from weedy rice plants at each of eight sampling sites **and** 1000 grains were collected from GM rice plants at each of eight sampling sites.

Plants were grown from the grains that had been collected and tested to determine the percentage gene flow between the weedy rice plants and the GM rice plants at each site.

Table 2.1 shows the results of the investigation.

Table 2.1

	percentage gene flow from GM rice plants to weedy rice plants	percentage gene flow from weedy rice plants to GM rice plants
	0.044	0.136
	0.013	0.136
	0.025	0.273
	0.019	0.235
	0.044	0.239
	0.025	0.242
	0.063	0.244
	0.057	0.273
mean:		0.222

- (i) Complete Table 2.1 by calculating the mean percentage gene flow from GM rice plants to weedy rice plants.

[1]

- (ii) The scientists decided to carry out a *t*-test to compare the percentage gene flow from GM rice plants to weedy rice plants with the percentage gene flow from weedy rice plants to GM rice plants.

The scientists stated the null hypothesis:

There is no difference between the percentage gene flow from GM rice plants to weedy rice plants and the percentage gene flow from weedy rice plants to GM rice plants.

The scientists calculated the value of *t* as **9.043**.

Table 2.2 shows the probability table for the t -test.

Table 2.2

degrees of freedom	critical values	
	$p = 0.05$ (5%)	$p = 0.01$ (1%)
12	2.179	3.055
13	2.160	3.012
14	2.145	2.977
15	2.131	2.947
16	2.120	2.921
17	2.110	2.898
18	2.101	2.878

Explain, with reference to Table 2.2, what can be concluded from the analysis of the data collected by the scientists.

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- (iii) The results of this investigation were published in a scientific paper.

A student who read the paper concluded that the results showed there were no reasons to be concerned about gene flow from GM plants to wild plants.

Suggest **one** reason why the results of this investigation should **not** be interpreted in this way.

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[Total: 10]